Audio and Speech Processing

EcAuTAL — Speech Processing

Marc Evrard

October, 2026



Section 1

Introduction to Speech Processing

Evolution of Speech Processing

- Digital speech processing studies speech signals and the digital methods applied to them (Rabiner and Schafer 2007)
- The field evolved from analog *vocoders* (Dudley 1939) to digital models in the 1960s–70s, then to machine learning
- Important milestones:
  - The Fast Fourier Transform (FFT, 1965)
  - Linear Predictive Coding (LPC, late 1960s)
  - Hidden Markov models for recognition (1980s)
  - Large self-supervised models (2020s)

Major Application Areas

- **Speech Coding:** Bandwidth compression for telecommunications
- **Speech Synthesis:** Converting text to natural-sounding speech (TTS)
- **Speech Recognition:** Transcribing speech into text (ASR or STT)
- **Speaker Verification:** Biometric security based on voice characteristics
- **Speech Enhancement:** Noise reduction and reverberation cancellation

Digital Representation of Speech

- Analog audio must be sampled and quantized before digital processing
- Typical sampling rates range from 8 kHz (telephony) to 48 kHz
- Many speech datasets use 16 kHz (ASR) or 22.05–24 kHz (TTS)
- Common sample formats are 16- or 24-bit linear PCM and 32-bit floating point

Challenges in Speech Processing

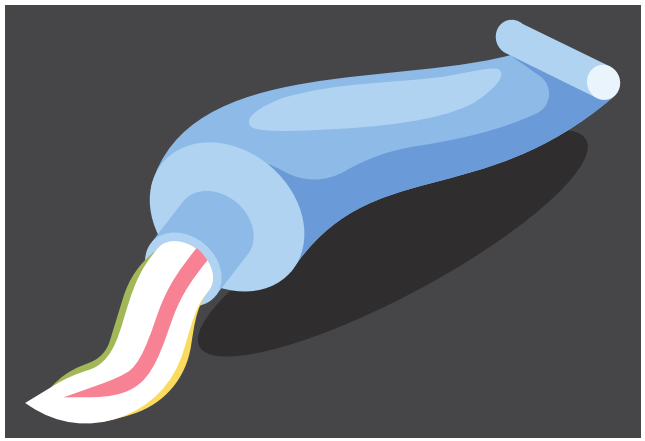
- **Variability:** Speech differs by speaker, accent, emotion, and context
- **Environment:** Background noise and acoustic reflections interfere with the signal
- **Real-Time Constraints:** Many applications need minimal latency
- **Information Overload:** Speech carries linguistic, paralinguistic, and personal information at once

Current Landscape of Speech Technology

- This course focuses on Text-to-Speech Synthesis (TTS) and Speech-to-Text (STT)
- The field moved from Statistical Parametric Speech Synthesis (SPSS) with HMMs to Deep Neural Networks (DNNs)
- End-to-End (E2E) architectures now dominate and minimize manual feature engineering
- They rely heavily on large datasets and high-performance computing (GPUs, TPUs)
- Self-supervised learning (SSL) models are increasingly integrated

Information Flow: STT vs. TTS

- **Speech-to-Text (STT):**
  - A ***many-to-one*** mapping problem
  - It reduces a continuous, highly variable acoustic signal to discrete linguistic units (graphemes or phonemes)
- **Text-to-Speech (TTS):**
  - A ***one-to-many*** mapping problem
  - It expands discrete text into a continuous waveform with endless possible variations (e.g., prosody, timbre)



Evaluation Metrics

- **STT:**
  - Primarily objective metrics:
    - Word Error Rate (WER)
    - Character Error Rate (CER)
- **TTS:**
  - Mostly subjective listening tests, such as the Mean Opinion Score (MOS)
  - Objective measures:
    - Mel-Cepstral Distortion (MCD, with DTW) and f_0 RMSE
    - ASR-based WER (intelligibility) and speaker similarity
    - MOS predictors (e.g., UTMOS)

The STT Pipeline

1. **Feature Extraction:** Converting raw audio into spectral frames
2. **Encoder:** Transforming features into high-level hidden representations
3. **Decoder/CTC:** Mapping hidden states to the final character or word sequence

The TTS Pipeline

1. **Text Frontend:** Normalization and Grapheme-to-Phoneme (G2P) conversion
2. **Acoustic Model:** Mapping phonemes to intermediate acoustic representations (e.g., Mel spectrograms)
3. **Vocoder:** Reconstructing the time-domain waveform from these features

Why Representation Matters

- Raw audio $x[n]$ is sampled at high rates, which produces long time series
- Time-domain signals are highly redundant and do not show frequency components explicitly
- Efficient representations add a **perceptual bias** that focuses the model on what the ear perceives

Outline

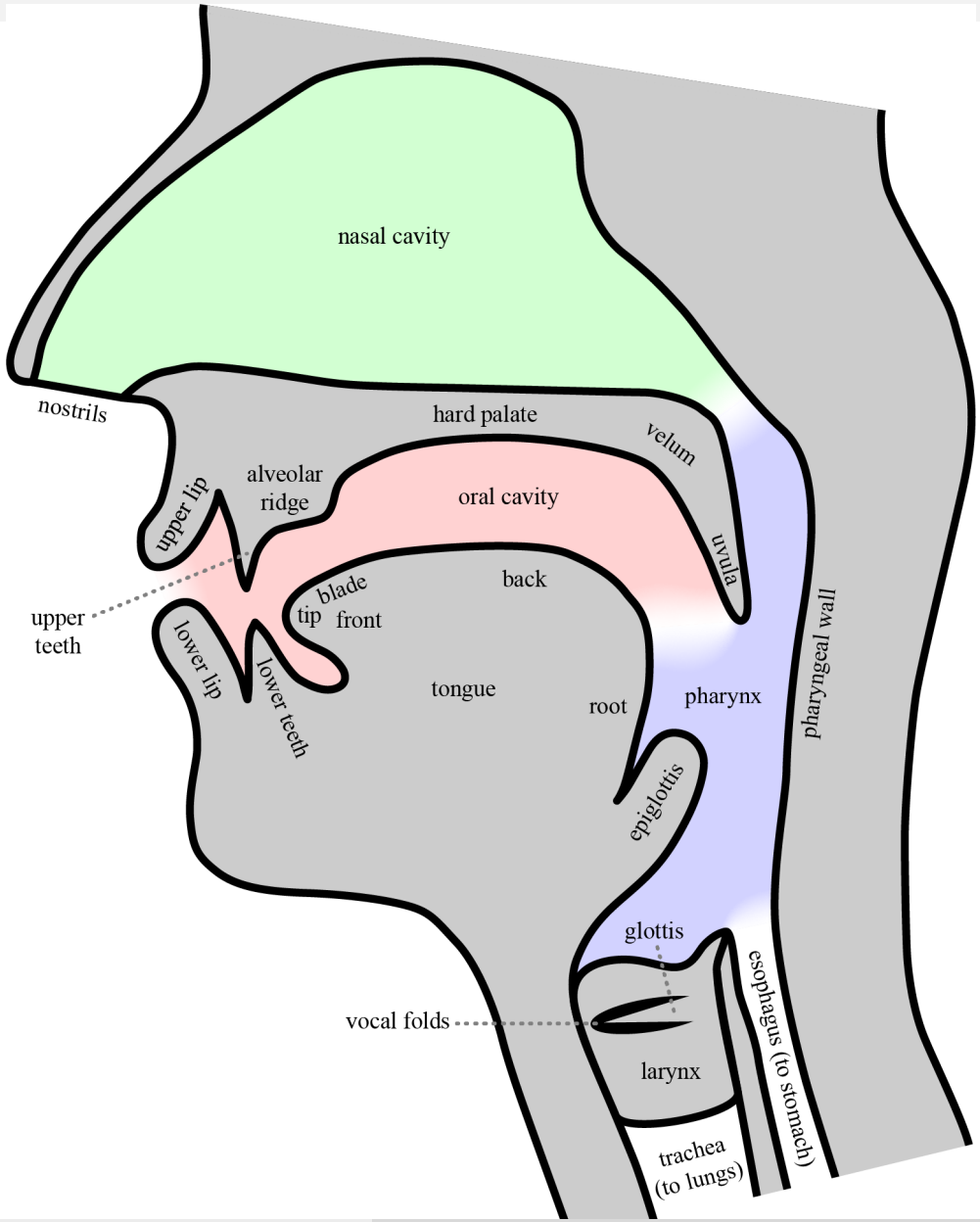
1. Introduction to Speech Processing
2. Fundamentals of Human Speech Production
3. Digital Audio Representation
4. Spectral and Parametric Representations
5. Linear Predictive Coding (LPC)
6. Signal Reconstruction

Section 2

Fundamentals of Human Speech Production

The Speech Production Mechanism

- The lungs are the power source and expel the air
- The air passes through the trachea into the larynx, which contains the vocal folds
- The vocal tract (pharynx, oral cavity, and nasal cavity) acts as a time-varying acoustic filter
- Articulators (tongue, lips, jaw, velum) change the vocal tract shape



(https://pressbooks.nvcc.edu/eng200h5p/)

A Speech Signal

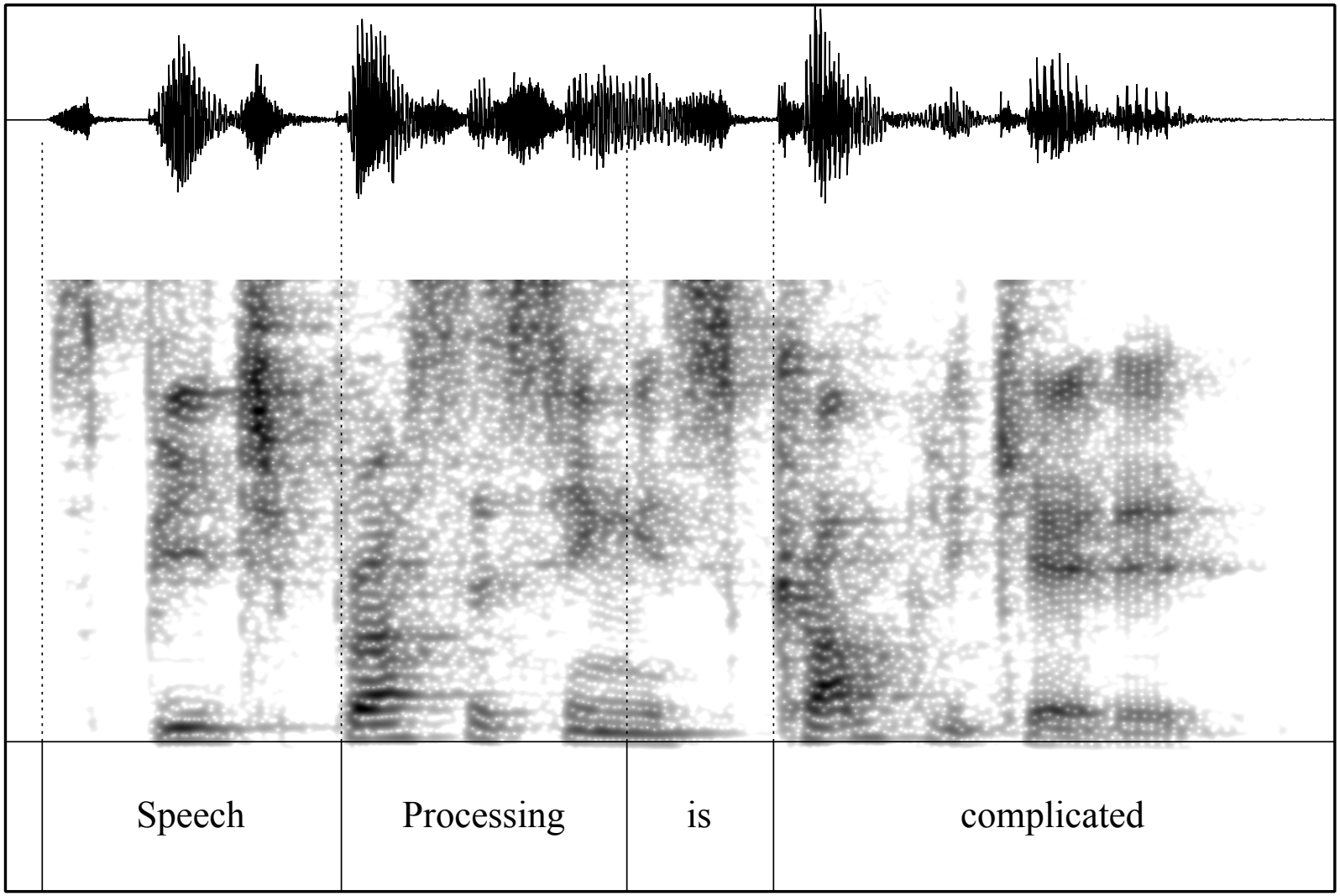


Figure 1: Praat visualization of a speech signal

Phonemes

- **Voiced Speech:** Vocal folds vibrate, producing quasi-periodic glottal pulses (e.g., vowels /a/, /u/)
- **Unvoiced Speech:** Vocal folds stay open, and air turbulence forms at a constriction (e.g., fricatives /s/, /f/)
- **Mixed:** Voiced fricatives combine both (e.g., /z/, /v/)
- **Plosives:** Pressure builds behind a closure and is suddenly released (e.g., /p/, /t/, /k/, or voiced /b/, /d/, /g/)

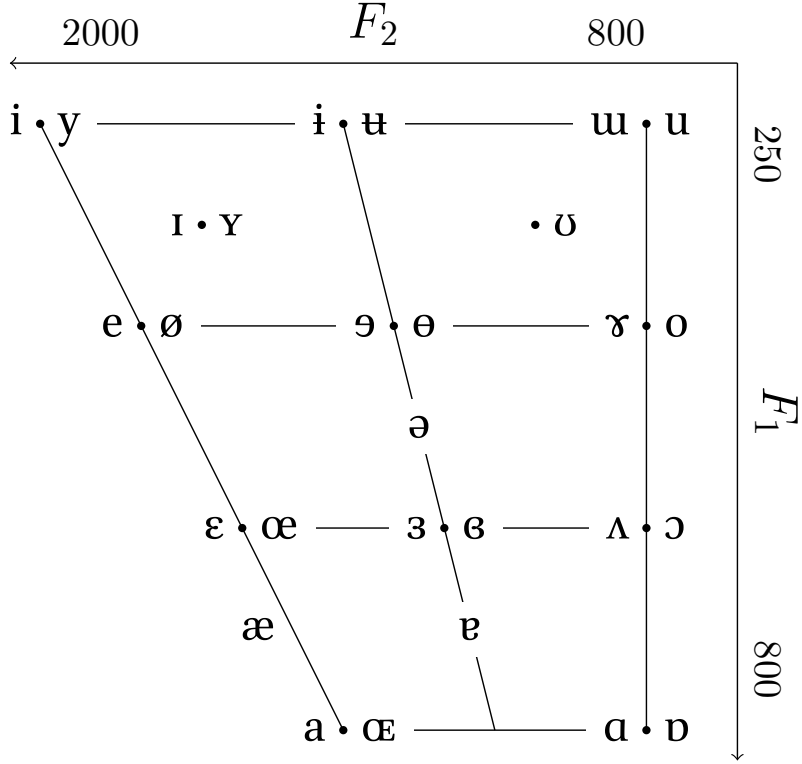


Figure 2: Vowel diagram with formants.

Vowels and Formants

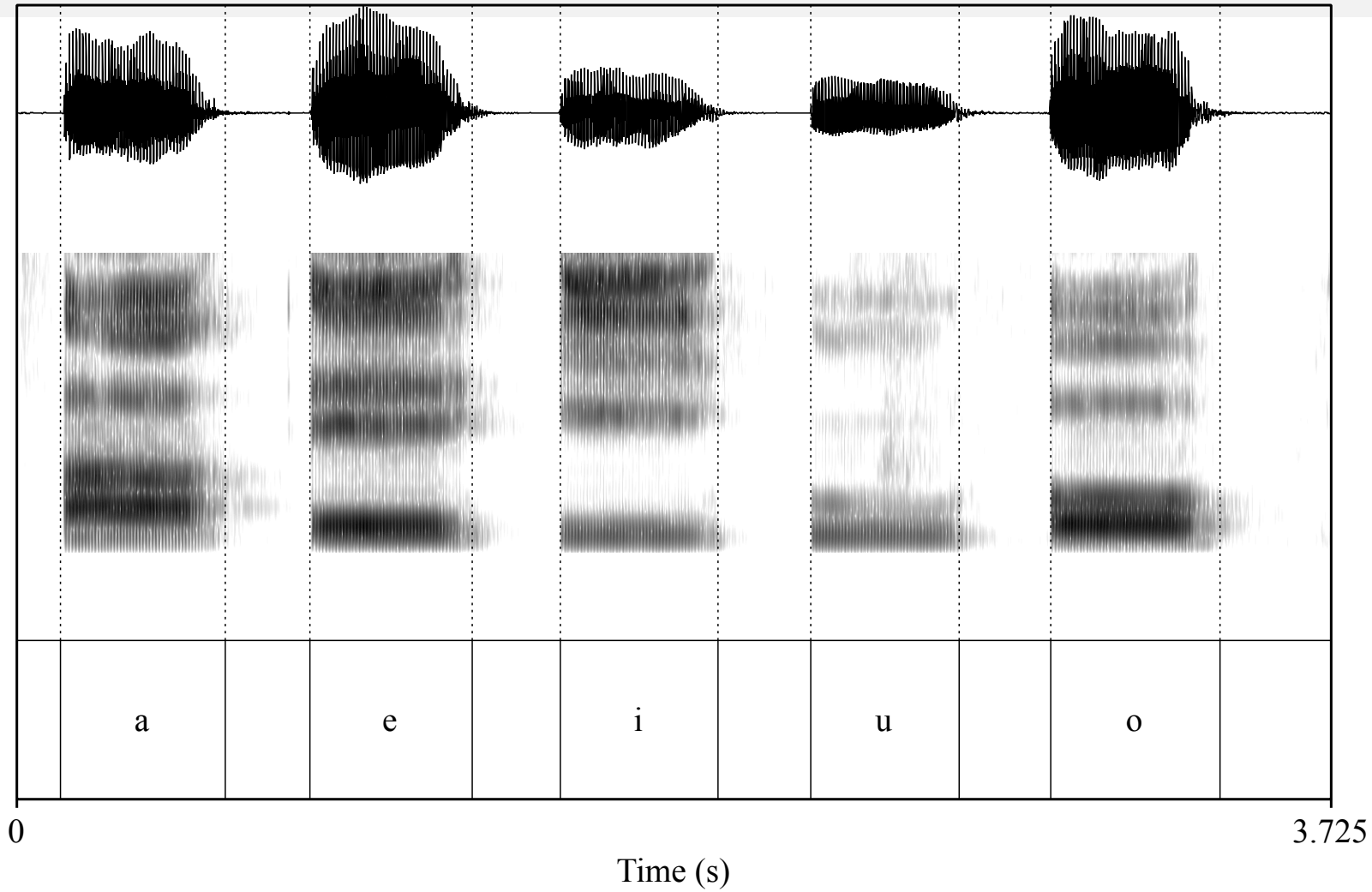


Figure 3: Vowel spectrograms.

Formant Minimal Pairs

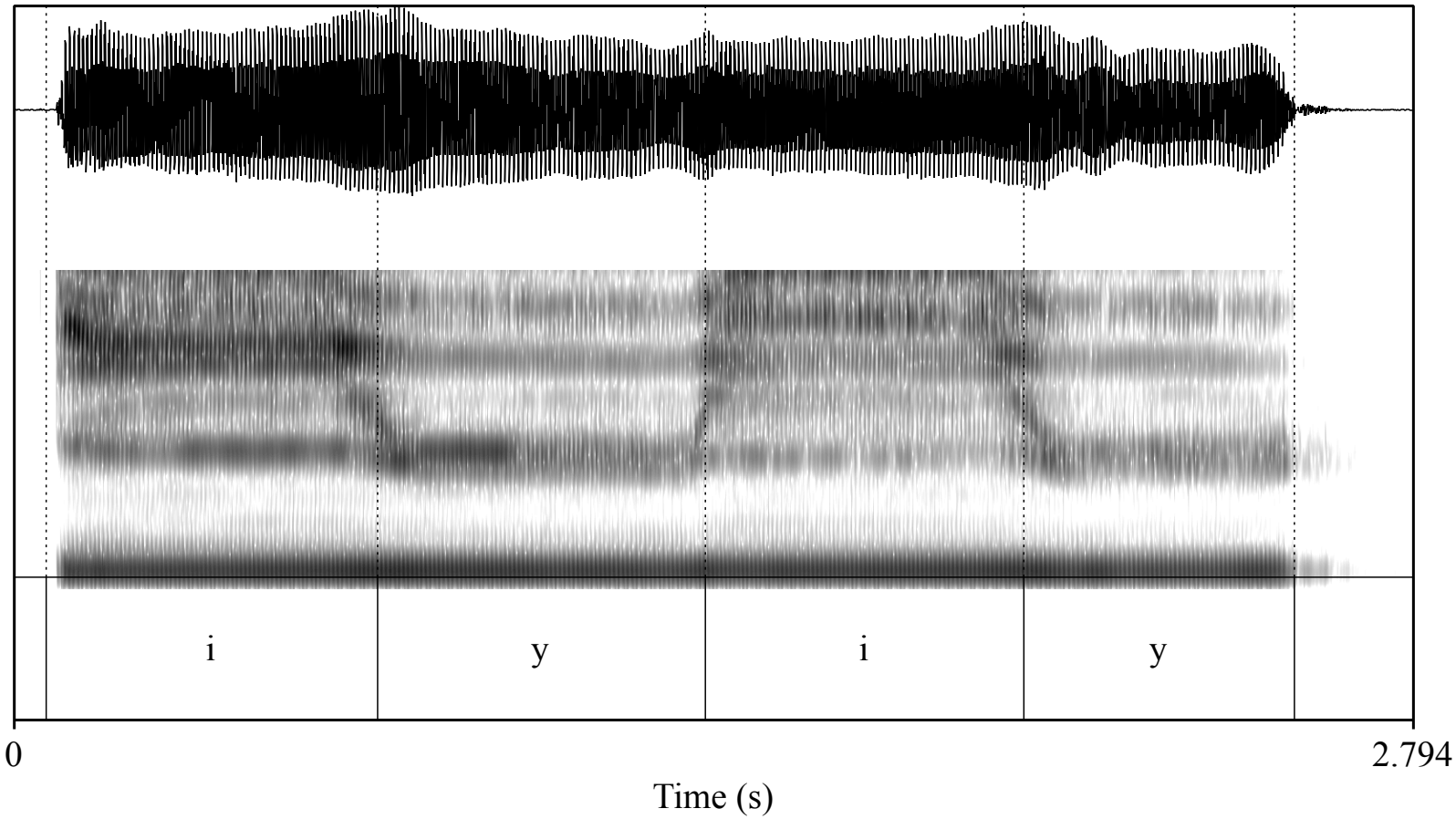


Figure 4: Spectrograms of vowels with two lip settings.

Fundamental Frequency (f_0)

- The rate at which the vocal folds vibrate is the fundamental frequency, f_0
- Perceptually, it corresponds to the *pitch* of the voice
- f_0 varies with sex, age, and intonation (e.g., ~100–120 Hz for men, ~200–220 Hz for women)

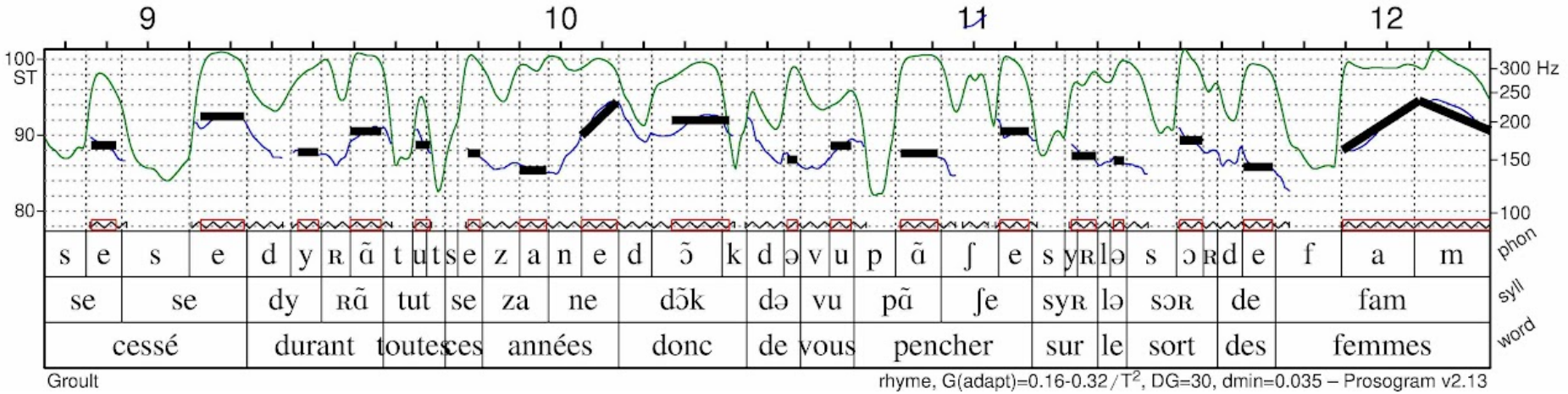


Figure 5: From https://sites.google.com/site/prosogram/home

The Source-Filter Model

- Fant (1960) formalized speech as a source signal passed through a linear filter
- It separates the excitation (source) from the vocal tract (filter)
- The source $e[n]$ represents the glottal pulses or noise
- The filter $h[n]$ represents the vocal tract resonances, called **formants**
- Lip radiation $r[n]$ converts airflow into pressure ($+6$ dB/octave)

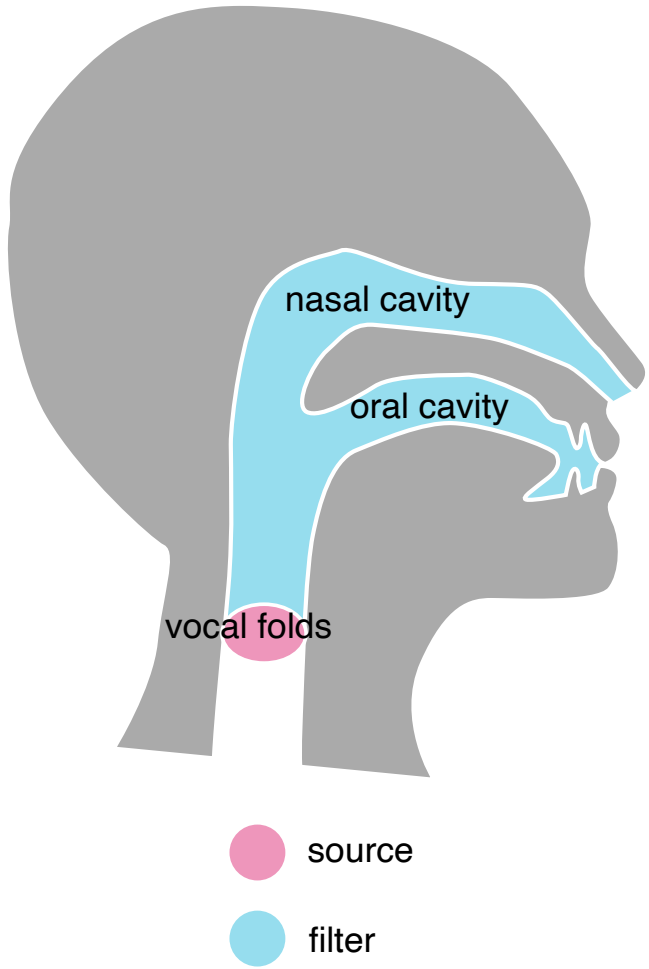


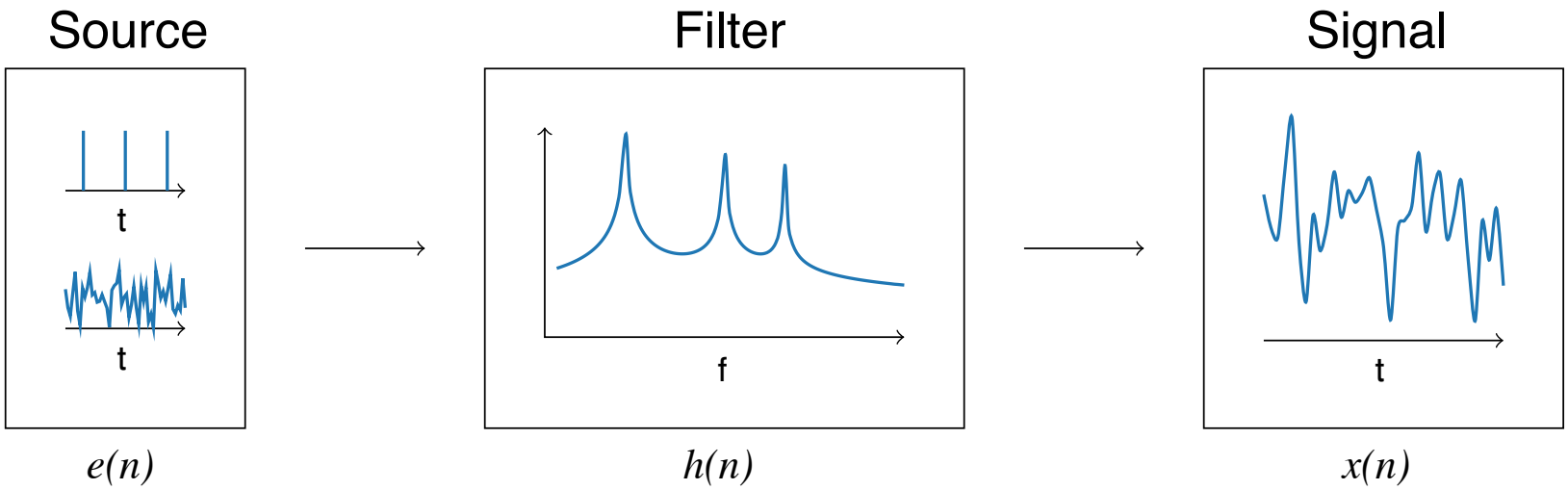
Figure 6: From Tokuda et al. (2013).

Transfer Function

- The speech signal $x[n]$ is the convolution of the components[†]:

$$x[n] = e[n] * h[n] * r[n] \quad \text{or} \quad X(z) = E(z) \cdot H(z) \cdot R(z)$$

- Vowels are mainly defined by the formants F_1, F_2, and F_3
- An all-pole vocal tract model suits non-nasal vowels well



[†] $X(z)$ is the Z-transform of $x[n]$ (Section 5)

Discrete-Time Model of Production

- The source is an impulse train for voiced phonemes and white noise for unvoiced ones
- A time-varying digital filter with impulse response $h[n]$ models the vocal tract resonances
- This model is the foundation of LPC

Biological Factors

- **Vocal Fold Dimorphism:** Larger vocal folds give men a lower f_0 than women
- **Aging (Presbyphonia):** Vocal fold atrophy or laryngeal ossification increases breathiness and reduces pitch stability
- **Hormonal Influence:** Testosterone and estrogen affect the hydration and thickness of the vocal fold mucosa, and thus vocal quality and range (e.g., at puberty)
- **Vocal Tract Scaling:** Formant frequencies are inversely proportional to the vocal tract length L ($F_n \approx \frac{(2n-1)c}{4L}$)†
- **Morphological Diversity:** Palatal shape, dental configuration, and nasal cavity volume influence sibilance and nasality

$^\dagger c$ is the speed of sound

Linguistic Variation

- **Diatopic:** Regional dialects and accents[†]
- **Diastratic:** Social class or education level
- **Diaphasic:** Situation and register (formal vs. casual)
- **Diachronic:** Language change over time
- **Diamesic:** Spoken vs. written medium

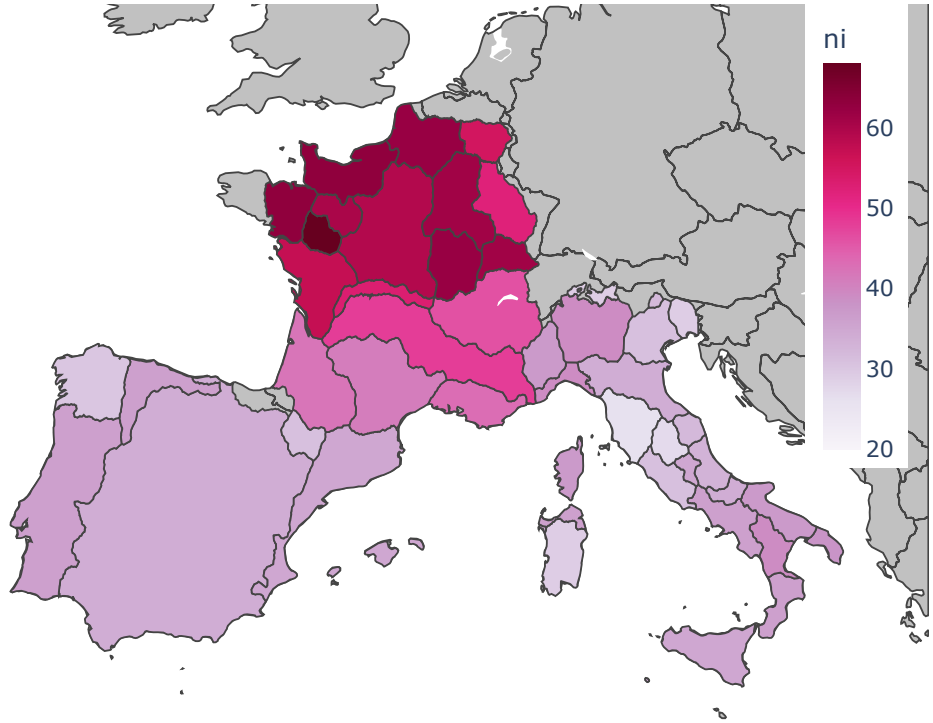


Figure 7: Romance dialect innovations relative to Latin (Boula de Mareüil et al. 2025).

[†] https://atlas.lisn.upsaclay.fr/?tab=EU

Frequency Selectivity

- The inner ear acts as a bank of overlapping band-pass filters, the critical bands (Zwicker and Terhardt 1980)

$$\Delta f_{CB} \approx 25 + 75 \left[1 + 1.4 \left(\frac{f}{1000} \right)^2 \right]^{0.69} \text{ Hz}$$

- This quasi-logarithmic resolution (~100 Hz below 500 Hz, 15–20% of f above 1 kHz) motivates the Bark (Traunmüller 1990) and Mel scales

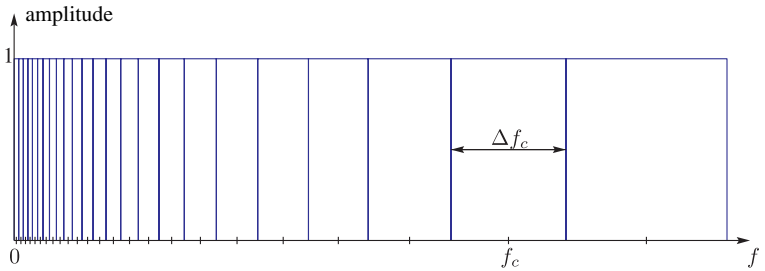


Figure 8: Critical band filters (Rabiner and Schafer 2007).

Perception of Intensity and Rhythm

- **Loudness:** Grows non-linearly with sound pressure (measured in phons and sones)
- **The Syllable:** Fundamental unit of speech rhythm and prosodic organization in most languages

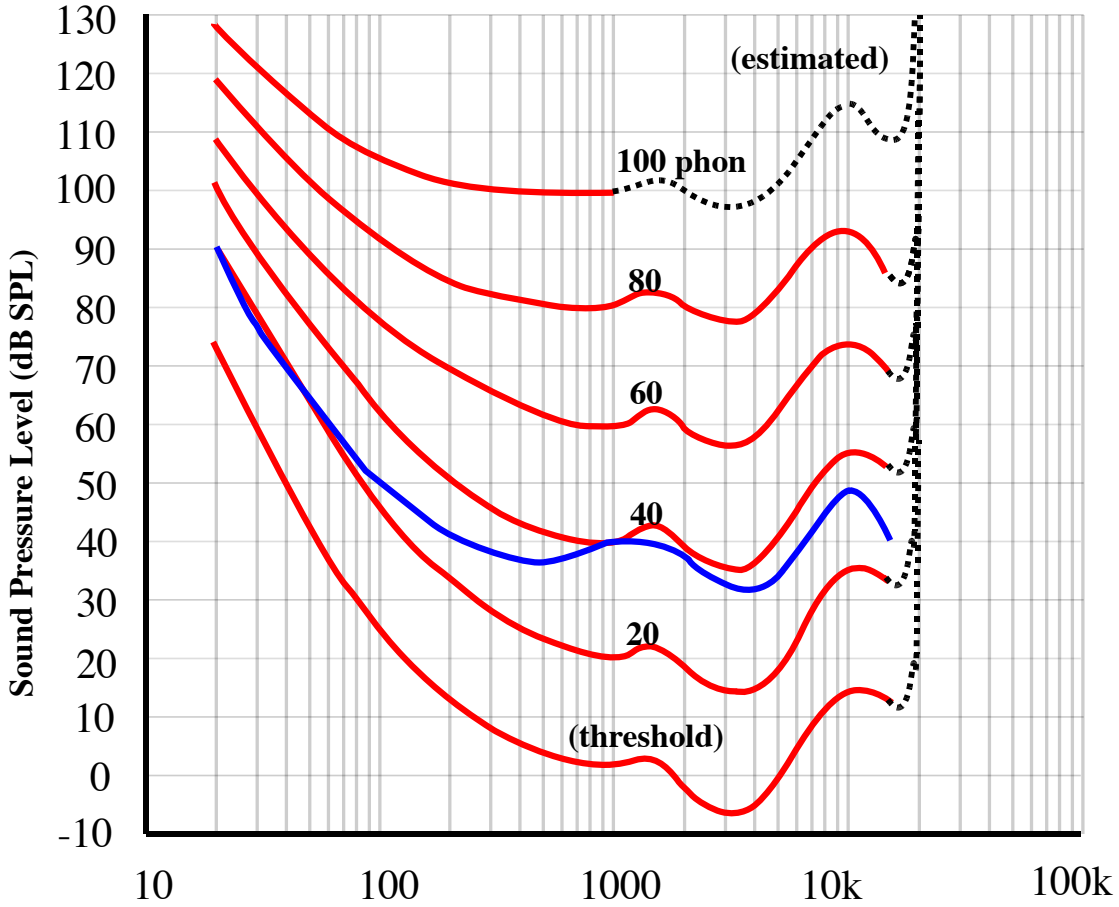


Figure 9: ISO equal-loudness contours, frequency in Hz (Wikimedia Commons).

Section 3

Digital Audio Representation

The Nyquist Limit and Analog Filters

- The Nyquist–Shannon sampling theorem requires a sampling rate $f_s > 2 \cdot f_{max}$ to prevent aliasing
- An analog *anti-aliasing* low-pass filter must remove all energy above $f_s/2$ before the A/D converter
- Sampling close to the Nyquist rate requires a *brick-wall* analog filter with a very steep roll-off
- Such ideal filters are physically impossible to build
- Practical approximations add severe phase distortion and high manufacturing costs

The Principle of Oversampling A/D

- The converter samples at a multiplied rate $f_{os} \gg 2 \cdot f_{max}$ (e.g., $64\times$ to $256\times$) but with a lower bit depth (e.g., 1 bit)
- **The Primary Benefit:** The analog anti-aliasing filter gets an extremely wide transition band
- A low-order RC filter then suffices, with nearly linear phase in the audio band
- A sharp, linear-phase *digital* filter removes everything above $f_s/2$ before downsampling (decimation)

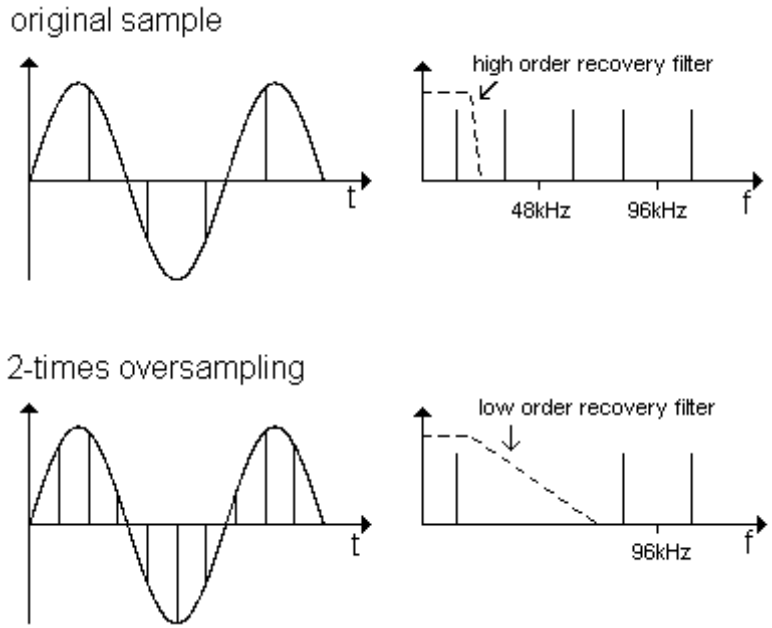


Figure 10: Oversampling vs. regular sampling, here for D/A. The reconstruction filter is relaxed the same way.

Sigma-Delta ($\Sigma\Delta$) Modulation

- Feedback loops inside the converter perform **noise shaping**
- Noise shaping needs spectral room above the audio band, so $\Sigma\Delta$ always relies on oversampling
- The quantization noise moves out of the audio band (low frequencies) into the high-frequency range, where the decimation filter removes it
- This lets fast 1-bit or low-resolution ADCs and DACs reach a 16- to 24-bit effective resolution
- A first-order modulator with quantization error $q[n]$ gives:

$$y[n] = x[n-1] + \underbrace{q[n] - q[n-1]}_{\text{high-pass shaped noise}}$$

Limitations of Fixed-Point (Integer) Audio

- Traditional Pulse Code Modulation (PCM) stores 16-bit (e.g., CD audio) or 24-bit fixed-point integers
- With B bits in two's complement, the hard maximum is $2^{B-1} - 1$ (bits $0111\ldots1$), e.g., $32\,767$ for 16 bits
- This maximum defines 0 dB_{FS}, and exceeding it causes **hard clipping**
- **Low-Level Resolution Loss:** Very quiet signals use fewer bits, so their relative quantization noise is higher

The Floating-Point Concept

- Floating point solves these issues with a binary scientific notation, $V = (-1)^S \cdot M \cdot 2^E$
- The bits are divided into three parts:
  1. **Sign bit (S):** Positive or negative
  2. **Mantissa/Significand (M):** The significant digits, which set the precision
  3. **Exponent (E):** A power-of-two multiplier that shifts the binary point
- Because the *binary point* floats, the *relative* precision is the same for very low and very high signals

The IEEE 754 Standard in Audio

- Modern 32-bit float audio formats follow the IEEE 754 single-precision standard

$$V = (-1)^{b_{31}} \times \left(1 + \sum_{i=1}^{23} b_{23-i} 2^{-i}\right) \times 2^{\left(\sum_{j=0}^{7} b_{j+23} 2^{j}\right) - 127}$$

- The 32 bits split into 1 sign bit, 8 exponent bits, and 23 mantissa bits
- The leading mantissa bit of a normal number is always 1 and is not stored, which gives **24 bits** of precision, as in 24-bit integer audio
- The 8-bit exponent spans many orders of magnitude

Headroom Flexibility

- A standard 24-bit fixed-point file has a theoretical dynamic range of about 144 dB
- 32-bit float has the same *precision* as 24-bit fixed point (24-bit mantissa), but relative to the signal, at any level
- Its *range* is far larger, with over 1500 dB between the smallest and largest normal values (2^{-126} and 2^{128})
- A 32-bit float signal can exceed 0 dB$_{FS}$, with values above 1.0 or below -1.0
- Audio that looks clipped in a Digital Audio Workstation (DAW) may still be intact and recoverable by lowering the volume

Advantages in Internal DSP and DAWs

- Almost all modern DAWs use 32-bit (or 64-bit) floating point in their mixing engines
- **Gain Staging Freedom:** A signal can be boosted by $+50$ dB, routed through a bus, and attenuated by -50 dB[†] on the master fader without clipping
- **Filter Stability:** Infinite Impulse Response (IIR) filters are highly sensitive to rounding errors (quantization noise and limit cycles)[†]

[†] While 50 dB does not result in a power-of-two multiplier, the resulting rounding error is negligible.

[†] To improve stability, complex filters are broken down into smaller, chained lower-order filters that stay stable despite rounding errors.

The Limitation of Standard ADCs

- Beyond internal DSP, 32-bit float can also improve recording hardware, which is bound by analog physics
- In a traditional system, the microphone signal passes through a single analog preamplifier and a single Analog-to-Digital Converter (ADC)
- An unexpectedly loud sound may overload the preamp or the ADC
- Recording a clipped signal into a 32-bit float file simply saves an accurate copy of a distorted wave

The Dual A/D Converter Architecture

- Modern 32-bit float field recorders reduce physical clipping by redesigning the input stage
- Instead of one ADC, the microphone signal is split into two parallel conversion paths
  - **Path 1 (High-Gain ADC):** Optimized for quiet sounds, it has a low noise floor but clips easily on loud transients
  - **Path 2 (Low-Gain ADC):** Optimized for loud events, it handles large voltages without overloading but has a higher noise floor

The Recording Process

- Inside the recorder, a dedicated DSP monitors both ADC streams in real time
- While the audio is quiet to moderately loud, the DSP records the High-Gain ADC
- When the audio approaches the clipping threshold of the High-Gain ADC, the DSP seamlessly crossfades to the Low-Gain ADC
- The combined stream is stored in the mantissa-and-exponent format of 32-bit floating point
- Because the exponent covers both small and large values, the final `.wav` file holds the full dynamic range of both converters

The End of Input Gain Setting

- In traditional recording, a sound engineer must meticulously *gain stage*, predicting the loudest sound that might occur and setting the input trim knob so peaks hit around -12 dBFS
- If the engineer guesses incorrectly, the recording is either permanently clipped (too loud) or buried in preamp hiss (too quiet)
- On 32-bit float recorders, the input gain knob has usually been removed, since the dual-ADC system captures dynamic ranges exceeding human hearing (typically over $130\,\text{dB}$ of physical range)
- In post-production, a whispered dialogue and a sudden larger acoustic event in the same audio file can be normalized independently

Section 4

Spectral and Parametric Representations

The Concept of Quasi-Stationarity

- Speech is **non-stationary**
- Over 10–30 ms intervals, however, it can be considered quasi-stationary
- Short-time analysis splits the signal into ***frames***
- Frames usually overlap, for continuity and reconstruction

Windowing Functions

- **Rectangular Window:** Its abrupt cutoff causes spectral leakage
- **Hamming/Hann Windows:** Tapered edges reduce discontinuities
- Window *length* sets the time–frequency trade-off, while its *shape* trades main-lobe width against side-lobe leakage

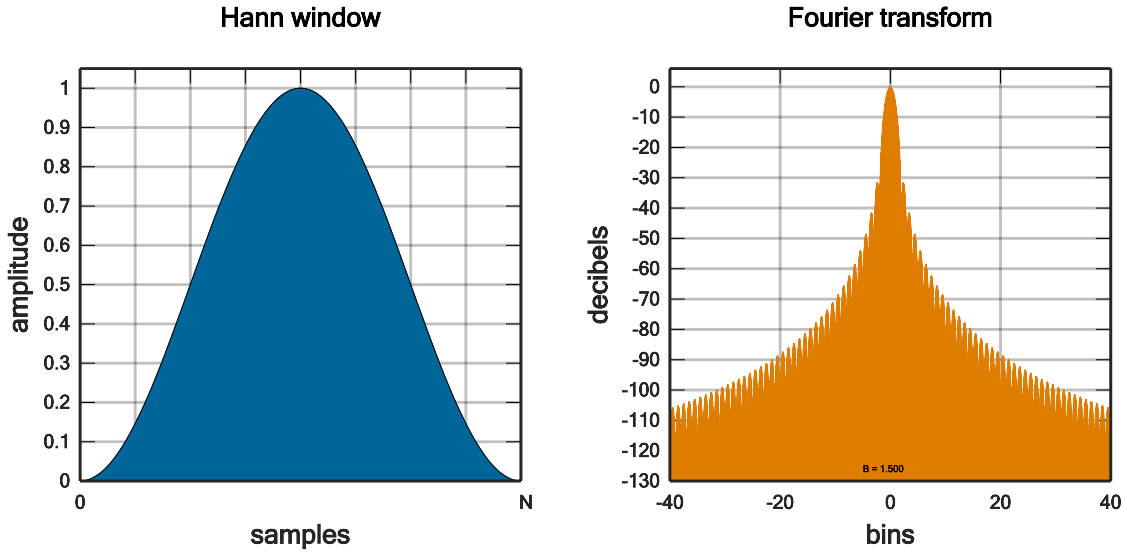


Figure 11: Hann window and its Fourier transform (Wikimedia Commons).

Pitch Period Estimation

- Autocorrelation measures the self-similarity of a frame (window $w[m]$ of length L) at different lags k:

$$R_n[k] = \sum_{m=0}^{L-1-k} \left(x[n+m]\, w[m]\right) \left(x[n+m+k]\, w[m+k]\right)$$

- For voiced (periodic) speech, the first major peak after the origin lies at the pitch period $T_0 = 1/f_0$
- This method tracks pitch robustly, even with some noise

The Discrete-Time Fourier Transform (DTFT)

- The DTFT of a sequence $x[n]$ is a *continuous*, 2π-periodic function of frequency:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

- $\omega = 2\pi f / f_s$ is the normalized angular frequency (rad/sample), and $\omega = \pi$ corresponds to the Nyquist frequency $f_s/2$
- **Inverse DTFT:**

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

- The DFT samples the DTFT of a length-N sequence at N equally spaced frequencies, $X[k] = X(e^{j\omega})\big|_{\omega = 2\pi k/N}$

The DFT and IDFT Equations

- The N-point Discrete Fourier Transform (DFT) maps a finite-length sequence $x[n]$ to a discrete spectrum $X[k]$
- **Forward DFT:**

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \ldots, N-1$$

- **Inverse DFT (IDFT):**

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \ldots, N-1$$

Important Properties of the DFT

- **Linearity:** $ax_1[n] + bx_2[n] \leftrightarrow aX_1[k] + bX_2[k]$
- **Periodicity:** Both $x[n]$ and $X[k]$ are implicitly periodic with period N
- **Circular Shift:** A time shift causes a linear phase shift in the frequency domain:

$$x[(n-m)_N] \leftrightarrow X[k]e^{-j\frac{2\pi}{N}km}$$

The Unit Impulse (Kronecker Delta)

- **Time Domain:** An impulse at the origin, $x[n] = \delta[n]$
- **Frequency Domain:** All frequencies, equally weighted, with zero phase

$$X[k] = 1, \quad \text{for all } k$$

- **Physical Meaning:** A single-sample click (like an idealized glottal pulse) spreads its energy evenly over the whole spectrum

The Delayed Impulse

- **Time Domain:** An impulse delayed by n_0 samples, $x[n] = \delta[n - n_0]$
- **Frequency Domain:** The magnitude stays constant, but the phase changes linearly

$$X[k] = e^{-j\frac{2\pi}{N}kn_0}$$

- **Physical Meaning:** A delay does not change the spectral energy distribution, only the relative timing (phase) of the frequency components

The Complex Exponential

- **Time Domain:** A complex exponential rotating at bin r, $x[n] = e^{j\frac{2\pi}{N}rn}$
- **Frequency Domain:** An impulse at bin r

$$X[k] = N\delta[k - r]$$

- **Physical Meaning:** It is the basis function of the Fourier transform

  It represents a pure, single-frequency component

The Real Cosine Wave

- **Time Domain:** A real-valued cosine at bin r, $x[n] = \cos\left(\frac{2\pi}{N} r n\right)$
- **Frequency Domain:** By Euler's formula, the energy splits equally into positive and negative (mirrored) frequencies

$$X[k] = \frac{N}{2}\left(\delta[k-r] + \delta[k-(N-r)]\right)$$

- **Physical Meaning:** Real-valued signals have conjugate-symmetric spectra, $X[N-k] = X^*[k]$, so their magnitudes are symmetric

  Because the DFT is N-periodic, the negative frequency $-r$ appears at bin $N-r$, in the upper half of the DFT

The Rectangular Window

- **Time Domain:** A pulse of L ones ($L \leq N$) padded with zeros, so $x[n] = 1$ for $0 \leq n < L$ and 0 otherwise
- **Frequency Domain:** A digital sinc function, formally the **Dirichlet kernel**

$$X[k] = e^{-j\frac{\pi k(L-1)}{N}} \frac{\sin\left(\frac{\pi k L}{N}\right)}{\sin\left(\frac{\pi k}{N}\right)}$$

- **Physical Meaning:** The abrupt edges in time create widespread *ripples* in frequency

  This is the *spectral leakage* seen when short speech frames are extracted without a smooth window

Frequency-Domain Filter Design

- **The Circular Convolution Problem:** Filtering directly in the frequency domain ($Y[k] = X[k] \cdot H[k]$) yields a circular convolution in the time domain
- **The Zero-Padding Solution:** For a signal block of length L and a filter of length M, circular and linear convolutions match if $N \geq L + M - 1$
- Ideal filters (e.g., a brick-wall low-pass) have an infinitely long impulse response (a sinc), which must be approximated by a finite one (FIR) of length M
- The ideal response is truncated to M samples, multiplied by a smooth window (e.g., Hamming or Hann), and delayed by $(M-1)/2$ samples to make it causal

Filter Banks

- A filter bank splits the signal into adjacent frequency bands, with one band-pass filter per channel
- Praat's *Filter (pass Hann band)* works in the frequency domain and multiplies the spectrum by a pass band with smooth, sine-shaped edges

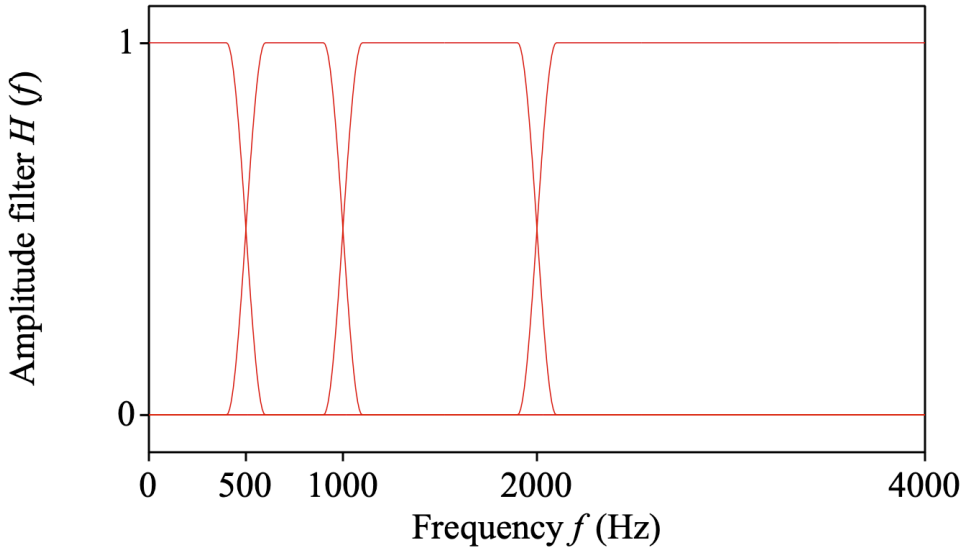


Figure 12: Three band-pass filters made with Praat's Hann band.

Definition of the STFT

- The Short-Time Fourier Transform (STFT) represents a signal in both time and frequency:

$$X(n, \omega) = \sum_{m=-\infty}^{\infty} x[m]\, w[n-m]\, e^{-j\omega m}$$

- It is the DTFT of the windowed speech segment around time n
- The output is a complex-valued function of time n and frequency ω
- In practice, an N-point FFT is computed every R samples (the hop size)
- Frame m starts at $n = mR$, and bin k has frequency $\omega_k = 2\pi k/N$, which gives the spectrogram $X(m, \omega_k)$

Linear Frequency vs. Mel Scale

- Digital Fourier analysis uses a linear frequency scale
- Human hearing, however, perceives frequency non-linearly
- The Mel scale and Bark scale are often used to warp the frequency axis for better perceptual modeling

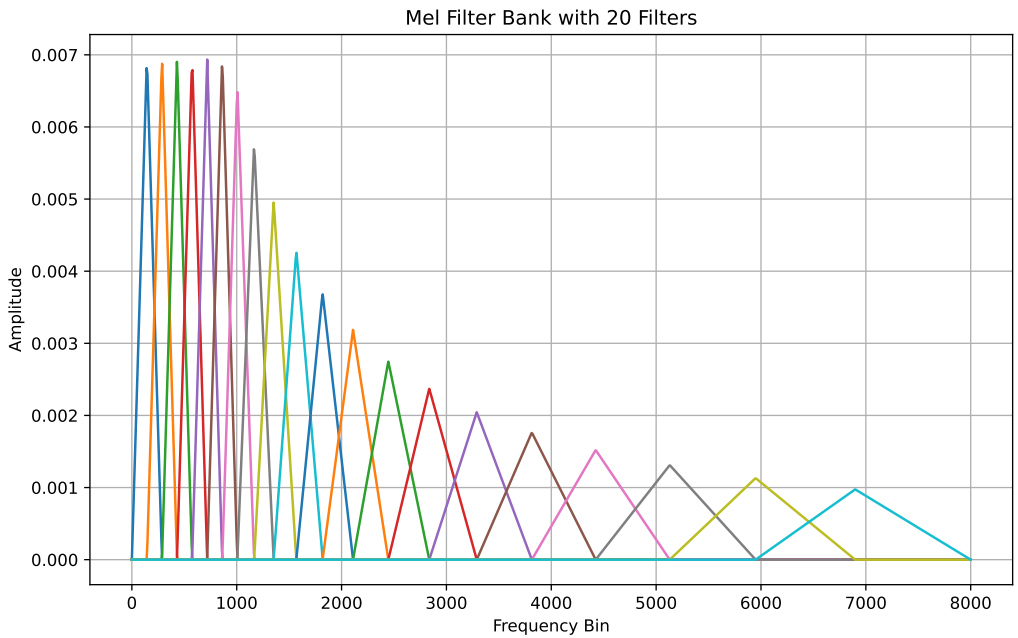


Figure 13: Mel-filter bank

Time-Frequency Trade-off: Narrowband

- **Narrowband Spectrogram:** Uses a long window (~20–30 ms)
- High frequency resolution (shows harmonics), poor time resolution

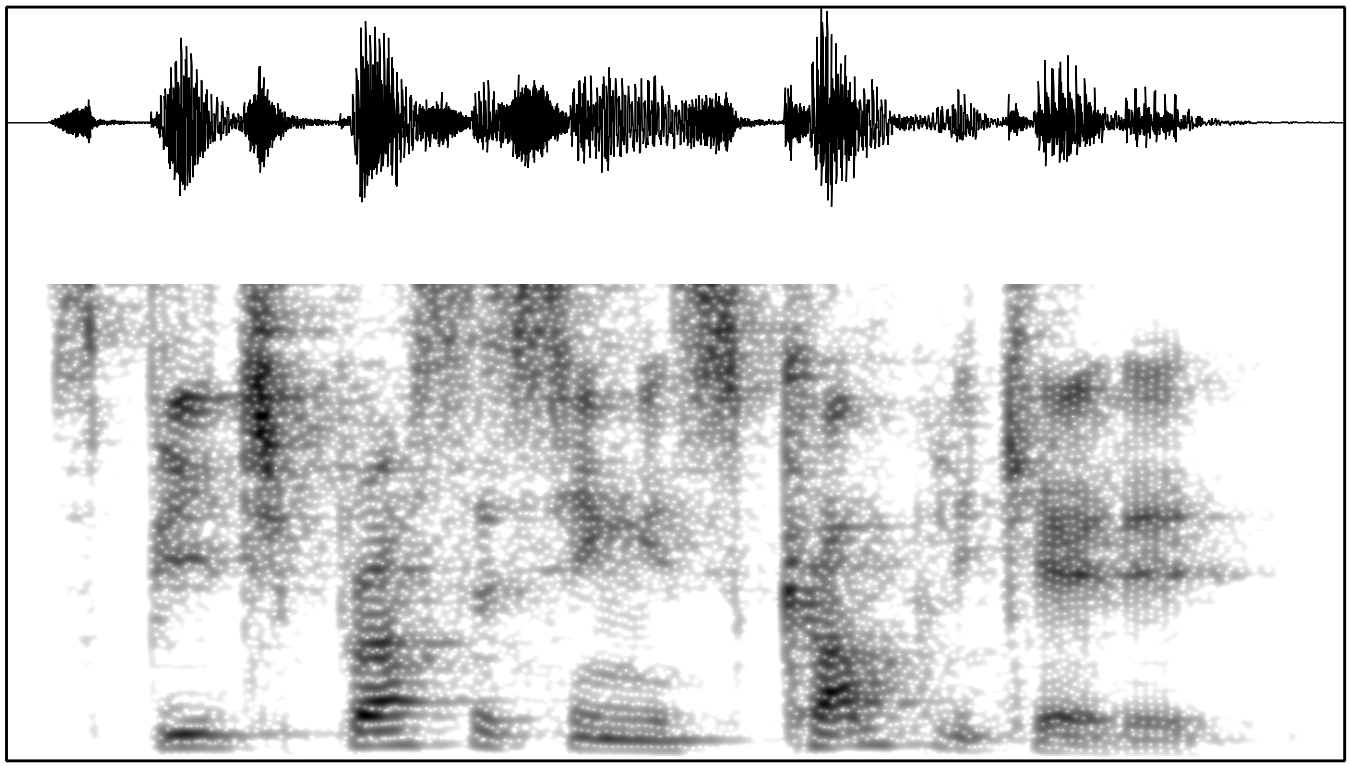


Figure 14: Spectrogram with 25 ms windows.

Time-Frequency Trade-off: Wideband

- **Wideband Spectrogram:** Uses a short window (~3–5 ms)
- High time resolution (one striation per glottal pulse, visible when zoomed in), poor frequency resolution (formants only)

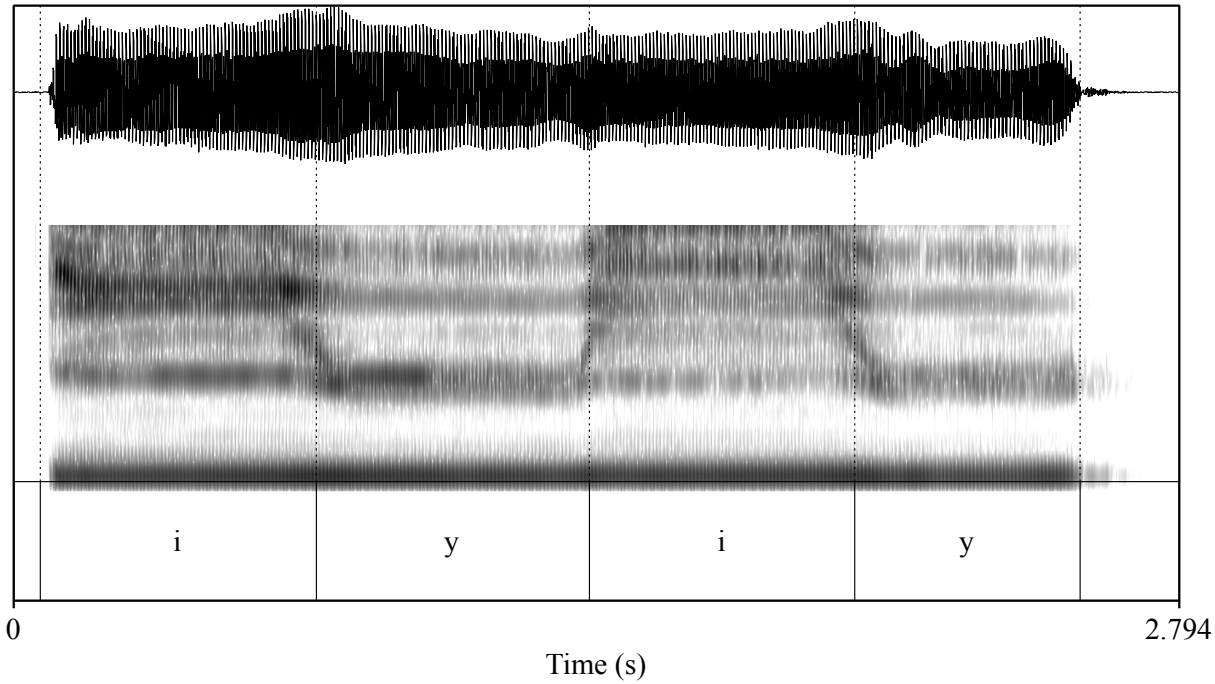


Figure 15: Spectrogram with 5 ms windows.

Homomorphic Systems

- A class of systems designed to separate convolved signals
- Since speech is $e[n] * h[n]$, we want to turn the convolution into an addition
- Taking the logarithm of the spectrum achieves this
- **Cepstrum:** The inverse Fourier transform of the log-magnitude spectrum

$$c[n] = \mathcal{F}^{-1}\{\log|\mathcal{F}\{x[n]\}|\}$$

- A Discrete Cosine Transform (DCT) sometimes replaces the inverse transform
- The x-axis of the cepstrum is called ***quefrency***

Separation of Source and Filter

- Along frequency, the vocal tract envelope varies slowly (low quefrency), while the harmonics ripple fast, every f_0 (a peak at quefrency $T_0 = 1/f_0$)
- By *liftering* (filtering in the cepstral domain), we can estimate the spectral envelope or the pitch independently

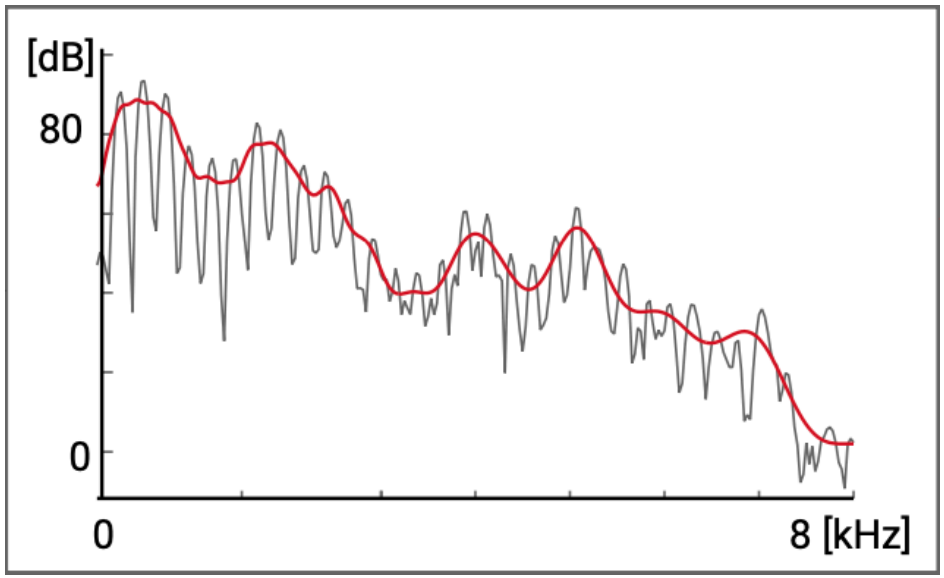


Figure 16: Log spectrum and its cepstrally smoothed envelope (red).

Delta Features

- Speech is dynamic, and static frames miss transitions
- **Delta (Δ):** First-order derivative of the coefficients over time
- **Delta-Delta ($\Delta\Delta$):** Second-order derivative (acceleration)
- Together, they capture the trajectory of spectral changes

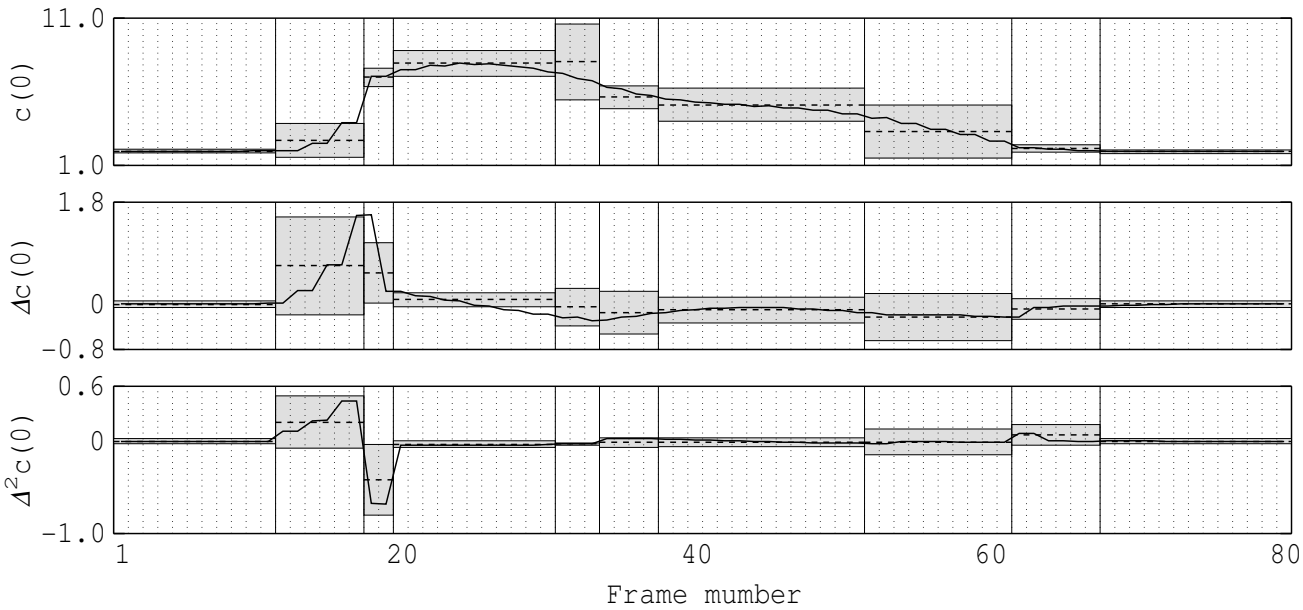


Figure 17: Static, delta, and delta-delta $c(0)$ generated by an HMM (Tokuda et al. 2013).

Section 5

Linear Predictive Coding (LPC)

From Time to the Complex Plane

- The DTFT (Section 4.3) evaluates signals only on the frequency axis (steady state)
- The Z-transform generalizes it to the whole complex plane
- It is the discrete-time counterpart of the Laplace transform, which models continuous-time analog circuits

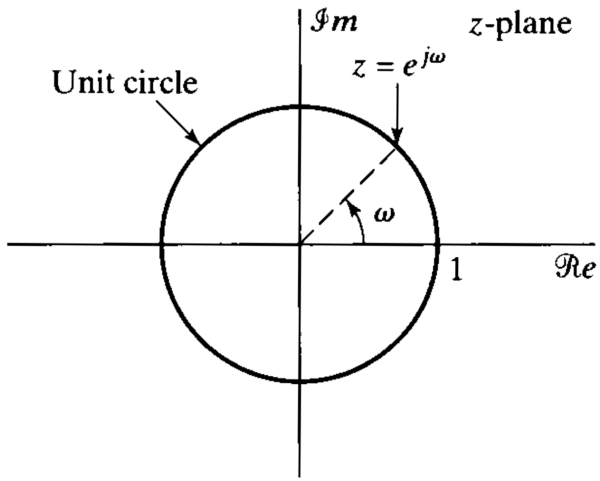


Figure 18: From Oppenheim, Schafer, and Buck (1999)

Definition

- The Z-transform of a discrete-time signal $x[n]$ is:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

- z is a complex variable, which can be written in polar form as $z = re^{j\omega}$
- r is the magnitude (distance from the origin), and ω is the angular frequency
- On the unit circle ($r = 1$), it reduces to the DTFT if the region of convergence (the values of z for which the sum is finite) includes the unit circle

The Transfer Function $H(z)$

- For a Linear Time-Invariant (LTI) system, the transfer function $H(z)$ is the ratio of the output $Y(z)$ to the input $X(z)$
- It is usually a ratio of polynomials in z^{-1} (delays):

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{Q} b_k z^{-k}}{1 - \sum_{k=1}^{P} a_k z^{-k}}$$

- Recursive digital filters use $Q + 1$ feedforward coefficients b_k (zeros) and P feedback coefficients a_k (poles)

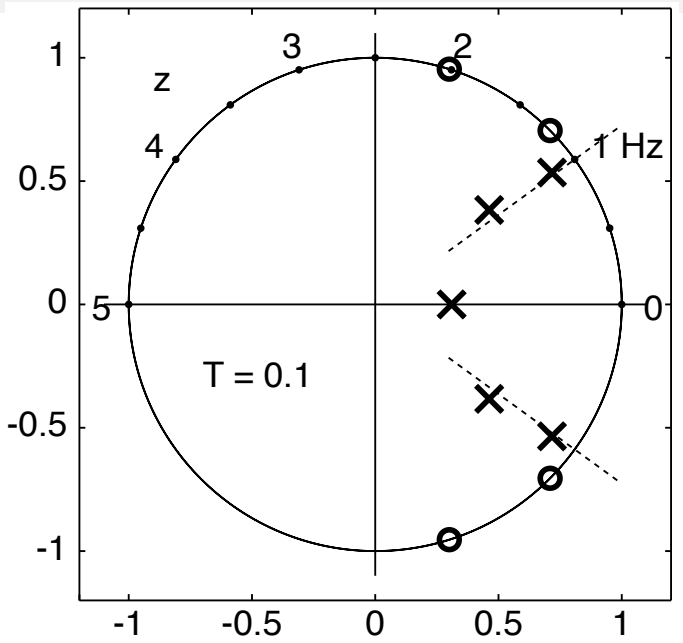
Poles and Zeros

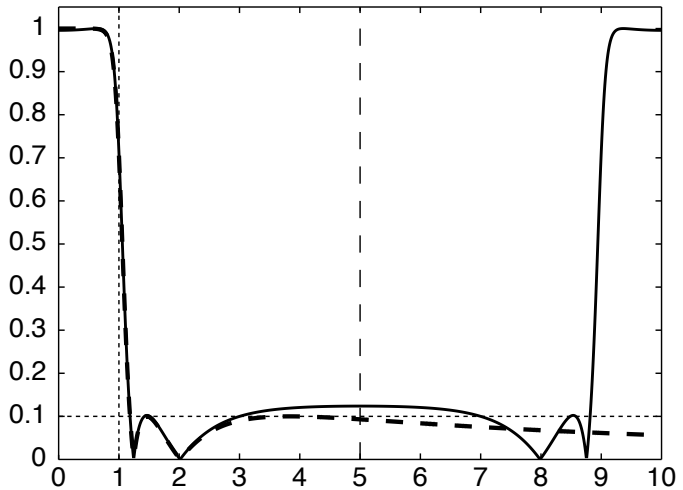
- **Zeros:** The complex roots of the numerator polynomial $(H(z) = 0)$

  $|H(e^{j\omega})|$ dips where the unit circle passes near a zero (exactly zero for a zero on the circle)

- **Poles:** The complex roots of the denominator polynomial $(|H(z)| \to \infty)$

  $|H(e^{j\omega})|$ peaks (resonance) where the unit circle passes near a pole





(Wikimedia Commons)

Stability and the Unit Circle

- The **unit circle** ($z = e^{j\omega}$) carries the frequency response
- From $\omega = 0$ to π, it spans 0 Hz to the Nyquist frequency $f_s/2$
- The distances from $e^{j\omega}$ to the zeros z_k and the poles p_k set the magnitude response:

$$|H(e^{j\omega})| = |b_0| \frac{\prod_k |e^{j\omega} - z_k|}{\prod_k |e^{j\omega} - p_k|}$$

- **Stability:** A causal filter is stable (its impulse response decays) if and only if all its poles lie strictly inside the unit circle ($|p_k| < 1$)

Modeling the Vocal Tract (LPC)

- In speech processing, LPC models the vocal tract as an ***all-pole*** system
- Each resonance of the mouth and throat (**formant**) corresponds to a complex-conjugate pair of poles inside the unit circle
- The closer a pole lies to the unit circle, the sharper the formant peak (narrower bandwidth)

The Linear Prediction Equation

- LPC predicts each speech sample $x[n]$ as a linear combination of past samples:

$$\tilde{x}[n] = \sum_{k=1}^{p} a_k x[n-k]$$

- p is the prediction order (typically 10–16 for speech)
- The a_k are the predictor coefficients

The Prediction Error (Residual)

- The prediction error is the difference between the actual and predicted samples:

$$e[n] = x[n] - \tilde{x}[n]$$

- LPC analysis minimizes the total squared error over a frame
- The residual $e[n]$ carries information about the excitation source

All-Pole Modeling

- The LPC model corresponds to an all-pole transfer function:

$$H(z) = \frac{G}{1 - \sum_{k=1}^{p} a_k z^{-k}}$$

- This matches the acoustic theory of non-nasal vowels, where resonances are poles
- A higher order p approximates the zeros of nasals and fricatives
- The poles also absorb the glottal pulse shape and lip radiation (an overall spectral tilt)
- A **pre-emphasis** filter $x[n] - \alpha\, x[n-1]$ ($\alpha \approx 0.97$) is therefore usually applied first

Solving for the Coefficients

- Minimizing $E = \sum_n e^2[n]$ over the windowed frame ($\partial E / \partial a_i = 0$) gives the **normal equations** (autocorrelation method):

$$\sum_{k=1}^{p} a_k \, R[|i-k|] = R[i], \quad i = 1, \ldots, p$$

  where $R[k]$ is the short-time autocorrelation of Section 4.2

- The matrix is Toeplitz (constant diagonals), so the **Levinson–Durbin** recursion solves it in $O(p^2)$ operations instead of $O(p^3)$

- Its reflection coefficients satisfy $|k_i| < 1$, so the all-pole filter is stable

- The gain is $G = \sqrt{E_p}$, with $E_p = R[0] - \sum_{k=1}^{p} a_k R[k]$

Section 6

Signal Reconstruction

Magnitude vs. Phase in the STFT

- The STFT yields a complex-valued spectrogram:

$$X(m,\omega) = |X(m,\omega)|e^{j\phi(m,\omega)}$$

- In many speech applications (e.g., TTS, speech enhancement), acoustic models predict or modify only the magnitude spectrum $|X(m,\omega)|$
- The phase $\phi(m,\omega)$ is often discarded because it *wraps*, looks quasi-random, and is thus hard to model and predict

The Challenge of Waveform Resynthesis

- Listening to generated or modified speech requires a real time-domain waveform $x[n]$
- The inverse STFT (ISTFT) needs both magnitude and phase
- Zero or random phase with the target magnitude produces strong artifacts
  - Zero phase sounds buzzy, with a pulse in every frame
  - Random phase sounds rough and noisy, though usually still intelligible

The Griffin-Lim Algorithm (GLA)

- Proposed by Griffin and Lim (1984), the GLA is a deterministic, iterative algorithm that estimates the missing phase
- **Note:** Not every complex 2D matrix is a valid STFT of a real time-domain signal
- STFT frames overlap heavily (e.g., 75%), which creates strong redundancy
- The algorithm forces the reconstructed spectrogram to be consistent across the overlapping windows

Initialization and the Target Spectrogram

- The input is a target magnitude spectrogram $|Y(m, \omega)|$ to convert into audio
- **Initialization ($i = 0$):** Start with an arbitrary phase estimate $\phi_0(m, \omega)$, usually uniform random noise
- The initial complex spectrogram combines the target magnitude with this random phase:

$$X_0(m, \omega) = |Y(m, \omega)| e^{j\phi_0(m,\omega)}$$

The Iterative Process (Part 1)

- **Step 1 (ISTFT):** Project the complex spectrogram into the time domain to get a signal estimate $x_i[n]$

$$x_i[n] = \text{ISTFT}\{X_{i-1}(m, \omega)\}$$

- The ISTFT uses overlap-add synthesis

  Since X_{i-1} is inconsistent, the overlapping frames conflict

  Overlap-add then acts as a smoothing constraint

- **Step 2 (STFT):** Recompute the STFT of this smoothed signal

$$\hat{X}_i(m, \omega) = \text{STFT}\{x_i[n]\}$$

The Iterative Process (Part 2)

- After Step 2, both the magnitude and the phase of $\hat{X}_i(m,\omega)$ differ from the target
- **Step 3 (Update Phase):** Keep the **new** phase of this consistent STFT and combine it with the **original** target magnitude

$$X_i(m,\omega) = |Y(m,\omega)|\frac{\hat{X}_i(m,\omega)}{|\hat{X}_i(m,\omega)|} = |Y(m,\omega)|e^{j\angle\hat{X}_i(m,\omega)}$$

- **Repeat:** Iterate Steps 1 to 3

  With each iteration, the STFT magnitude of $x_i[n]$ gets closer to the target $|Y(m,\omega)|$

Limitations and Acoustic Artifacts

- **Convergence:** The squared error between the target and reconstructed magnitudes never increases, but the algorithm often gets stuck in local minima
- **Computational Cost:** Repeated FFTs and IFFTs ($\sim$50 to 100 iterations) make it expensive and hard to use in real-time, low-latency applications
- **Audio Quality:** GLA inherently adds a reverberant or robotic phase-smearing artifact
- Modern pipelines are therefore largely replacing GLA with **neural vocoders** (e.g., WaveNet, HiFi-GAN), which learn to synthesize phase implicitly from vast amounts of audio data

Bibliography I

Boula de Mareüil, Philippe et al. (2025). "Computer modelling of innovations relative to Latin in contemporary Romance dialects." In: *Isogloss. Open Journal of Romance Linguistics* 11.3, pp. 1–31.

Dudley, Homer (1939). "Remaking speech." In: *The Journal of the Acoustical Society of America* 11.2, pp. 169–177.

Fant, Gunnar (1960). *Acoustic theory of speech production: with calculations based on X-ray studies of Russian articulations*. The Hague: Mouton.

Griffin, Daniel and Jae Lim (1984). "Signal estimation from modified short-time Fourier transform." In: *IEEE Transactions on Acoustics, Speech, and Signal Processing* 32.2, pp. 236–243.

Oppenheim, Alan V., Ronald W. Schafer, and John R. Buck (1999). *Discrete-time signal processing*. 2nd ed. Upper Saddle River, NJ: Prentice Hall.

Bibliography II

Rabiner, Lawrence R. and Ronald W. Schafer (2007). "Introduction to digital speech processing." In: *Foundations and Trends® in Signal Processing* 1.1-2, pp. 1–194.

Tokuda, Keiichi et al. (2013). "Speech synthesis based on hidden Markov models." In: *Proceedings of the IEEE* 101.5, pp. 1234–1252.

Traunmüller, Hartmut (1990). "Analytical expressions for the tonotopic sensory scale." In: *The Journal of the Acoustical Society of America* 88.1, pp. 97–100.

Zwicker, Eberhard and Ernst Terhardt (1980). "Analytical expressions for critical-band rate and critical bandwidth as a function of frequency." In: *The Journal of the Acoustical Society of America* 68.5, pp. 1523–1525.